Climate

The big idea

India has a **monsoon type of climate**. The word *monsoon* comes from the Arabic **mausim**, meaning season. Winds reverse their direction with the seasons, and almost all of India's rainfall comes in a few months. Understanding the monsoon explains India's seasons, its farming calendar, and even its festivals.

WEATHER VS CLIMATE

Weather is the state of the atmosphere over an area at any point of time. **Climate** is the sum total of weather conditions and variations over a large area for a long period of time (more than **thirty years**). Elements of both: temperature, atmospheric pressure, wind, humidity and precipitation.

Variations within India

- **Temperature:** in summer the Rajasthan desert can touch **50°C**, while **Pahalgam** in Kashmir may be around 20°C. On a winter night **Drass** in Ladakh may drop to **minus 45°C**, while Thiruvananthapuram has about 22°C.
- **Precipitation:** upper Himalayas get snowfall; elsewhere rain. Annual rainfall ranges from over **400 cm in Meghalaya** to **less than 10 cm in Ladakh and western Rajasthan**. Most of India gets rain from **June to September**, but the **Tamil Nadu coast** gets much of its rain in **October and November**.
- Coastal areas have less contrast in temperature; interior areas have seasonal contrast (**continentality**).

Factors affecting climate (climatic controls)

1. **Latitude:** temperature falls from the Equator to the poles.
2. **Altitude:** temperature decreases as height increases; hills are cooler.
3. **Pressure and wind system:** these depend on latitude and altitude and influence temperature and rainfall.
4. **Distance from the sea (continentality):** the sea moderates climate; interior places have extreme climate.
5. **Ocean currents** and **relief features** also matter: high mountains block cold or hot winds and cause rain on the windward side.

Factors affecting India's climate:

- **Latitude:** the **Tropic of Cancer** passes through the middle; the south is tropical, the north sub-tropical.
- **Altitude:** the Himalayas (average height about 6,000 m) **prevent cold winds from Central Asia** entering India, so the subcontinent has milder winters.
- **Pressure and winds:** India lies in the region of the **north-easterly winds**. These blow from the subtropical high-pressure belt of the Northern Hemisphere. In summer, winds reverse, and the **south-west monsoon** winds blow.
- **Upper air circulation: westerly jet stream** (south of the Himalayas in winter) brings **western cyclonic disturbances** to the north-west in winter; an **easterly jet stream** blows over the Peninsula in summer at about 14°N.
- **Western cyclonic disturbances:** low-pressure systems from the Mediterranean region that bring winter rain to the north-west.
- **Tropical cyclones:** form in the Bay of Bengal and Arabian Sea in the monsoon and post-monsoon periods; affect the coasts.

JET STREAMS

Jet streams are a narrow belt of high-altitude (above 12,000 m) **westerly winds** in the troposphere. Their speed varies from about **110 km/h in summer to about 184 km/h in winter**.

The Indian monsoon

The monsoon is experienced in the tropical area roughly between **20°N and 20°S**. Factors that explain its mechanism:

1. **Differential heating and cooling of land and water** creates low pressure over the Indian landmass and high pressure over the surrounding seas.
2. The shift of the **Inter Tropical Convergence Zone (ITCZ)** in summer over the **Ganga plain** (this is the equatorial trough, normally about 5°N; also called the **monsoon trough** during the monsoon).
3. The presence of a **high-pressure area east of Madagascar**, about 20°S over the Indian Ocean; its intensity and position affect the Indian monsoon.
4. The **Tibetan plateau** gets intensely heated in summer, causing strong vertical air currents and low pressure over it at about 9 km above sea level.
5. The movement of the **westerly jet stream** to the north of the Himalayas and the presence of the **tropical easterly jet stream** over the Peninsula during summer.
6. **El Niño:** warm ocean currents that flow past the coast of Peru instead of the cold Peruvian current, every two to five years; associated with weaker monsoons.
7. **Southern Oscillation (SO):** changes in pressure between the Indian Ocean and the south Pacific. The difference between pressure at **Tahiti** (French Polynesia, Pacific) and **Darwin** (northern Australia) is used to predict the strength of the monsoon. Together these are called **ENSO**.

Onset and withdrawal: The monsoon arrives on the southern tip (Kerala) by about **1 June**. It splits into two branches: the **Arabian Sea branch** and the **Bay of Bengal branch**. The Arabian Sea branch reaches Mumbai about ten days later, around **10 June**. Both branches merge over the north-western part of the Ganga plains. Delhi gets monsoon showers from the Bay of Bengal branch by the end of June (about 29 June). By mid-July the monsoon covers the whole country (about **8 July**).

Withdrawal begins in the north-western states by early **September**; by mid-

October it withdraws from northern India completely, and from the Peninsula by early December.

Breaks in the monsoon: the monsoon has wet and dry spells; rain falls for a few days at a time, then dry spells (breaks) occur. These are linked with the movement of the monsoon trough. The monsoon is known for its **uncertainties**: it may cause floods in one place and drought in another; its onset and withdrawal may be irregular.

The seasons

India has four main seasons:

Season	Months	Key features
Cold weather (winter)	Mid-November to February	December and January coldest; north-east trade winds blow from land to sea, so most of India is dry; **Tamil Nadu coast gets rain** as these winds blow from sea to land there; north gets light winter rain from **western cyclonic disturbances**, locally called **mahawat**, very useful for **rabi crops**; snowfall in the Himalayas
Hot weather (summer)	March to May	Heat belt moves north; in May up to 45°C in the north-west; **loo**: strong, gusty, hot dry winds over north and north-west India; dust storms; **mango showers** (pre-monsoon rain in Kerala and Karnataka that help early ripening of mangoes); **Kaal Baisakhi** (calamity of the month of Baisakh): evening thunderstorms in West Bengal; **blossom showers** in Kerala (coffee flowers)
Advancing monsoon (rainy season)	June to September	South-west monsoon brings heavy rain; windward side of the Western Ghats gets over 250 cm; **Mawsynram** in the **Khasi hills**

Season	Months	Key features
		receives the highest average rainfall in the world; breaks in rainfall
Retreating/post-monsoon	October and November	Low-pressure trough weakens; skies clear; temperature rises; high temperature and humidity make weather oppressive: "**October heat**"; low-pressure conditions move to the Bay of Bengal, causing **cyclonic depressions** that strike the east coast (deltas of Godavari, Krishna and Kaveri); much rain in the Coromandel coast

TAMIL NADU'S RAIN

Tamil Nadu gets most of its rainfall in **winter (October–December)** from the retreating monsoon and north-east winds, not in June–September, because it lies in the rain-shadow of the Arabian Sea branch.

Distribution of rainfall

- **Heavy rainfall** (over 400 cm): western coast and north-east.
- **Less than 60 cm**: western Rajasthan and adjoining parts of Gujarat, Haryana and Punjab; also the interior of the Deccan plateau and east of the Sahyadris; Leh in Jammu and Kashmir.
- Rest of the country gets moderate rainfall. Snowfall is restricted to the Himalayan region.
- Rainfall is **highly variable** from year to year.

The monsoon as a unifying bond

The Himalayas protect the subcontinent from cold winds; the Peninsular plateau, under the influence of the sea, has a moderate climate. The entire agricultural

calendar, life of people and their festivals revolve around the monsoon. Year after year people wait eagerly for its arrival; the monsoon winds bind the whole country together by giving water for farming.

BOARD FAVOURITE

"Why is the Indian monsoon called a unifying bond?" and "Mention the factors that affect the mechanism of the monsoon" are frequently asked. Also: "Why does the Coromandel coast receive rain in winter?"

Quick recap

- Climate depends on latitude, altitude, pressure and winds, distance from sea, ocean currents and relief.
- Monsoon: differential heating, ITCZ shift, Tibetan plateau heating, jet streams, El Niño and Southern Oscillation.
- Onset in Kerala about 1 June; whole country by mid-July; withdrawal from September.
- Seasons: winter, summer (loo, mango showers, Kaal Baisakhi), rainy, retreating (October heat).
- Mawsynram gets the world's highest average rainfall; western Rajasthan very little.

Practice questions

1. Which place receives the highest average rainfall in the world?
   (a) Cherrapunji in the Western Ghats (b) Mawsynram (c) Thiruvananthapuram (d) Mumbai
2. Hot, dry and gusty winds blowing in north India during summer are called
   (a) mango showers (b) Kaal Baisakhi (c) loo (d) mahawat
3. The pre-monsoon showers that help the early ripening of mangoes in Kerala and Karnataka are called
   (a) mango showers (b) blossom showers (c) loo (d) October heat
4. Winter rainfall in north-west India is caused by
   (a) the south-west monsoon (b) tropical cyclones of the Bay of Bengal (c) the retreating monsoon (d) western cyclonic disturbances
5. Which region of India receives rainfall mainly in winter?
   (a) Punjab (b) Tamil Nadu coast (c) Kerala (d) Meghalaya
6. The Southern Oscillation compares pressure between
   (a) Mumbai and Chennai (b) Peru and Madagascar (c) Tahiti and Darwin (d) Tibet and Kerala
7. The monsoon usually arrives at the southern tip of India by
   (a) 15 May (b) 1 June (c) 10 June (d) 15 July
8. Describe the factors that affect the mechanism of the Indian monsoon.
9. Why is the Indian monsoon considered a unifying bond?

Answers: 1. b 2. c 3. a 4. d 5. b 6. c 7. b 8. see notes 9. see notes